

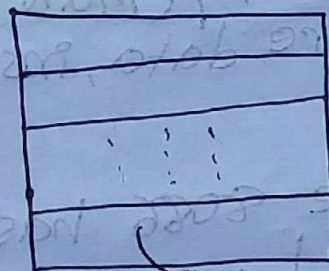
8086 memory segmentation:

- Total memory size is divided into segments of various sizes.
- A segment is just an area in memory.
- The process of dividing memory this way is called segmentation.

#

Address Bus : 20 bit

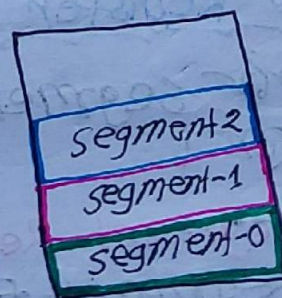
Data Bus : 16 bit



20 bit index size

#

segmentগুলোতে ডিফল্ট নাম রাখা হয়- code similar data ডিফল্ট নাম থাকে।



RAM

4 types of segment in 8086:

- i) code segment
- ii) Data segment
- iii) Stack segment
- iv) Extra segment.

Segment:

A segment is a continuous area of memory (max. 64 KB) that the 8086 uses to store data, instructions, or stack information.

The 8086 has:

$\left. \begin{array}{l} \rightarrow 20 \text{ bit address bus} \\ \rightarrow 16 \text{ bit Data Bus.} \end{array} \right\} \text{so 20 Register use 20 address \& \text{ sum}}$

Two Register in a segment:

Each segment is point by 20 Register लाइ।

i) segment Register: (16 bit)

segment ki Top ka sum index ka point dikhata hai।

ii) offset Register: (16 bit)

segment ki exact position ka pointer dikhata hai।

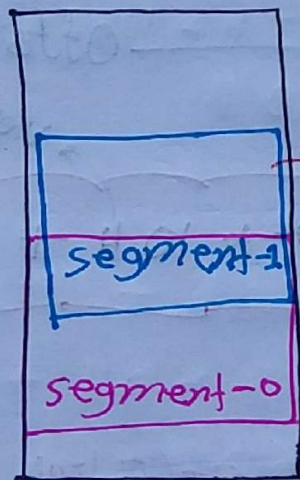
8086 4 segment 4 tr:

	Segment Register	Offset Register
i) Code segment	CS	IP
ii) Data segment	DS	BX, DI, SI
iii) Stack segment	SS	SP, BP
iv) Extra segment	ES	BX, DI, SI

Overlapping segment:

segment started before end of current segment:

যদি segment এর শুরু আগে কি বা বিবাদিত
নতুন segment started হতে পারে।



RAM

এই ব্যাপার ব্যবহার হলো (যেমন)
segment declare করে use না
করে পাঠানো হলে প্রোগ্রামার
কিছু সমস্যা না হয়।

offset 0 ચર્ચાની starting address માટે

Starting address of segment : $\text{segment} \times 10 \text{ H}$

ending address of segment : $\text{segment} \times 10 \text{ H} + \text{FFFF H}$

offset 255 16 bit ની છે offset ની Max^m value ~~છે~~ માટે ચર્ચાની નીચે address માટે ~~છે~~ **FFFF**

Ex^m $2222 \text{ H} + 0016 \text{ H} = 2222 \text{ H} + 16 \text{ H} = 2223 \text{ H}$

$$\begin{aligned}\text{Virtual address} &= \text{CS} : \text{IP} \\ &= 2222 \text{ H} : 0016 \text{ H}\end{aligned}$$

$$\begin{aligned}\text{Physical address} &= \text{segment} \times 10 \text{ H} + \text{offset} \\ &= \text{CS} \times 10 \text{ H} + \text{IP} \\ &= 2222 \text{ H} \times 10 \text{ H} + 0016 \text{ H} \\ &= 22220 \text{ H} + 0016 \text{ H} \\ &= 22236 \text{ H}\end{aligned}$$

Ex^m A memory location has physical address 805D2 H. In what segment does it have offset BFD2 H?

solⁿ

$$\begin{aligned}\text{Physical address} &= \text{segment} \times 10 \text{ H} + \text{offset} \\ \Rightarrow \text{segment} &= \frac{\text{physical address} - \text{offset}}{10 \text{ H}} \\ &= \frac{805\text{D}2 \text{ H} - \text{BFD}2 \text{ H}}{10 \text{ H}} = 7960 \text{ H}\end{aligned}$$

Ex: The contents of the following registers are:

CS = 1111 H, DS = 3333 H, SS = 2526 H.

IP = 1232 H, SP = 1100 H, DI = 0020 H

calculate the corresponding physical address

Soln

code segment

$$\text{Physical address} = \text{CS} \times 10\text{H} + \text{IP}$$

$$= 1111\text{H} \times 10\text{H} + 1232\text{H}$$

$$= 12342\text{H}$$

Data segment

$$\text{Physical address} = \text{DS} \times 10\text{H} + \text{DI}$$

$$= 3333\text{H} \times 10\text{H} + 0020\text{H}$$

$$= 33350\text{H}$$

Stack segment

$$\text{Physical address} = \text{SS} \times 10\text{H} + \text{SP}$$

$$= 2526\text{H} \times 10\text{H} + 1100\text{H}$$

$$= 26360\text{H}$$

Ans

$$26360\text{H} - 25260\text{H} = 1100\text{H}$$

$$26360\text{H} - 25260\text{H} = 1100\text{H}$$

$$26360\text{H} - 25260\text{H} = 1100\text{H}$$

Ex: 31

For the memory Location whose physical address is specified by 1256A H, given the address in segment : offset, for segments 1256 H, 1240 H

Solⁿ

For segment 1256 H.

$$\text{Physical address} = \text{segment} \times 10 \text{ H} + \text{offset}$$

$$\begin{aligned}\Rightarrow \text{Offset} &= \text{Physical address} - \text{segment} \times 10 \text{ H} \\ &= 1256 \text{ A H} - 1256 \text{ H} \times 10 \text{ H} \\ &= 1256 \text{ A H} - 12560 \text{ H} \\ &= 000 \text{ A H}\end{aligned}$$

$$\text{So, } 1256 \text{ A H} = 1256 \text{ H} : 000 \text{ A H}$$

For segment 1240 H

$$\text{Physical address} = \text{segment} \times 10 \text{ H} + \text{Offset}$$

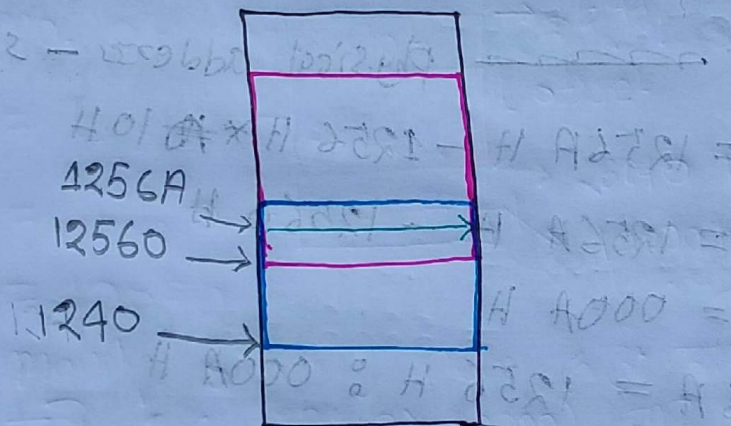
$$\begin{aligned}\text{offset} &= \text{Physical address} - \text{segment} \times 10 \text{ H} \\ &= 1256 \text{ A H} - 1240 \times 10 \text{ H} \\ &= 1256 \text{ A H} - 12400 \text{ H} \\ &= 016 \text{ A H}\end{aligned}$$

$$\text{So, } 1256 \text{ A H} = 1240 \text{ H} : 016 \text{ A H}$$

N.B

જ્યારે physical address ની ત્રીસ 26 virtual or Logical address ની overlapping ની થાય છે.

When overlapping happens, one physical address can have multiple Logical/virtual address.



Size of a segment

Starting index ની Last index ની 20 index ની size ની સુધારા (સાદાસાદા) segment size.

0000 $\longleftrightarrow$ FFFF

of indices in a segment = 2^{16}

$$\begin{aligned} \text{size of segment} &= 2^{16} \times 1B \\ &= \frac{2^{16} \times 1}{2^{10}} \text{ KB} \\ &= 64 \text{ KB} \end{aligned}$$

જ્યારે index ની size 1 Byte

N.B ৱাশে ৷ segment overlap হবে so
 সা সাইজ ৷ Block টি total size মোতর বহুলি
 কিন্তু process টি size মোতর নাহে না !

summer - 25 Q 3(a)

(I)

$$\begin{aligned}\text{Physical address} &= \text{SS} * 10\text{H} + \text{SP} \\ &= 7\text{CA}2\text{H} * 10\text{H} + 3344\text{H} \\ &= 7\text{FD}64\text{H}\end{aligned}$$

$$\therefore \text{PA} = \text{segment} * 10\text{H} + \text{offset}$$

$$\begin{aligned}\Rightarrow \text{segment} &= \frac{\text{PA} - \text{offset}}{10} \\ &= \frac{7\text{FD}64 - \text{B}294}{10} \\ &= 79\text{ADH}\end{aligned}$$

(II)

$$\begin{aligned}\text{Physical Address} &= \text{DS} * 10\text{H} + \text{SI} \\ &= 6\text{D}2\text{A} * 10\text{H} + 11\text{AA} \\ &= 6\text{E}44\text{AH}\end{aligned}$$

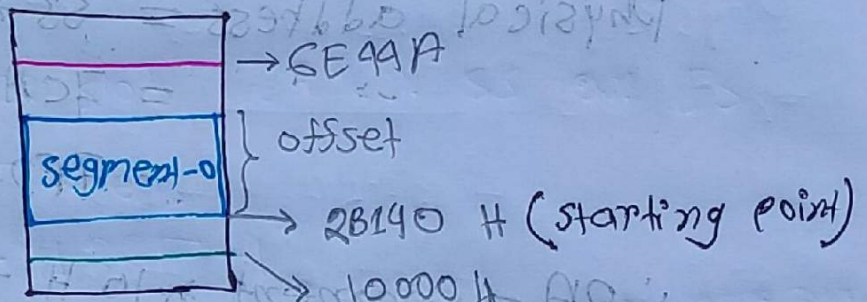
$$\text{Physical Address} = \text{segment} * 10\text{H} + \text{offset}$$

$$\begin{aligned}\text{offset} &= \text{PA} - \text{segment} * 10\text{H} \\ &= 6\text{E}44\text{A} - 2\text{B}41 * 10 \\ &= 4330\text{AH}\end{aligned}$$

invalid because offset is 16 bit that is $FFFF > \dots$, But here $FFFF < 9330H$

main ~~সমস্যা~~ ~~২৫~~ ~~২৫~~ offset ~~২৫~~ 16 bit ~~২৫~~ But ~~২৫~~ ২০ bit ~~২৫~~ ~~২৫~~!

Visualization



mainly starting point ~~২৫~~ যদি physical address ~~২৫~~ $(2B140 > 10000)$ যা ending point ~~২৫~~ physical address ~~২৫~~ $(6E44A > 2B140)$ ~~২৫~~ invalid

Ⓘ

Spring 25 Q3(a)

code segment:

$$\text{starting index} = \text{segment} \times 10H$$

$$= 25A1 \times 10$$

$$= 25A10H$$

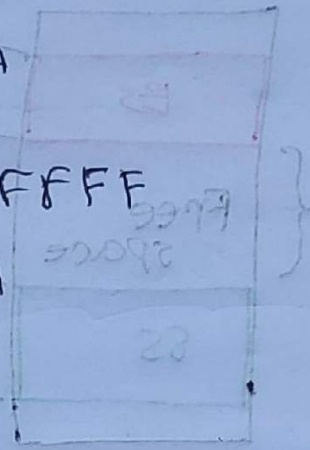
$$\text{ending index} = 25A10 + FFFF$$

$$= 35A0F$$

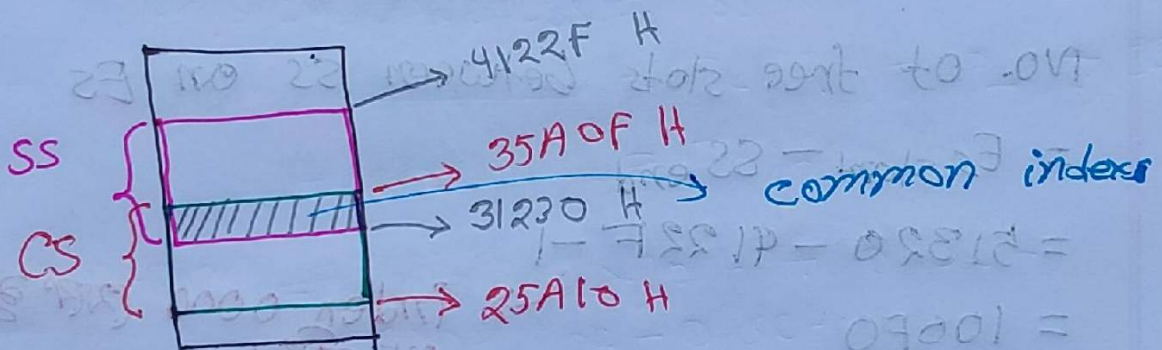
stack segment:

Starting index: $3123 \times 10H$
 $= 31230H$

ending index: $31230 + FFFF$
 $= 4122FH$



~~No. of overlapping slots =~~



No. of overlapping slots = $(35A0FH - 31230H) / 10H + 1$

$= 47E0H$
 $= 18400$ (Decimal)

II

stack segment:

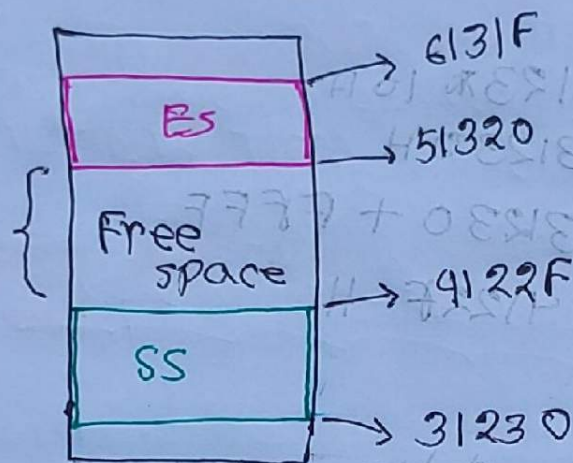
Starting: $5132 \times 10H$
 $= 51320H$

ending = $51320H + FFFF$
 $= 6131FH$

extra segment:

Starting: $5132 \times 10H$
 $= 51320H$

ending = $51320H + FFFF$
 $= 6131FH$



No. of free slots between SS and ES

$$\begin{aligned}
 &= ES_{start} - SS_{end} \\
 &= 51320 - 4122F - 1 \\
 &= 100F0
 \end{aligned}$$

index 0000 (মানে শুরু ২য় স্লট)

So, Free slots $> FFFF + 1$

index 0000 (মানে শুরু ২য় স্লট)

So place var with overlapping var

Fall 25 Q3(a)

CS

$$\begin{aligned}
 \text{Starting} &: 35E9 * 104 \\
 &= 35E90
 \end{aligned}$$

$$\begin{aligned}
 \text{ending} &: 35E90 + FFFF \\
 &= 45E8F
 \end{aligned}$$

Ds

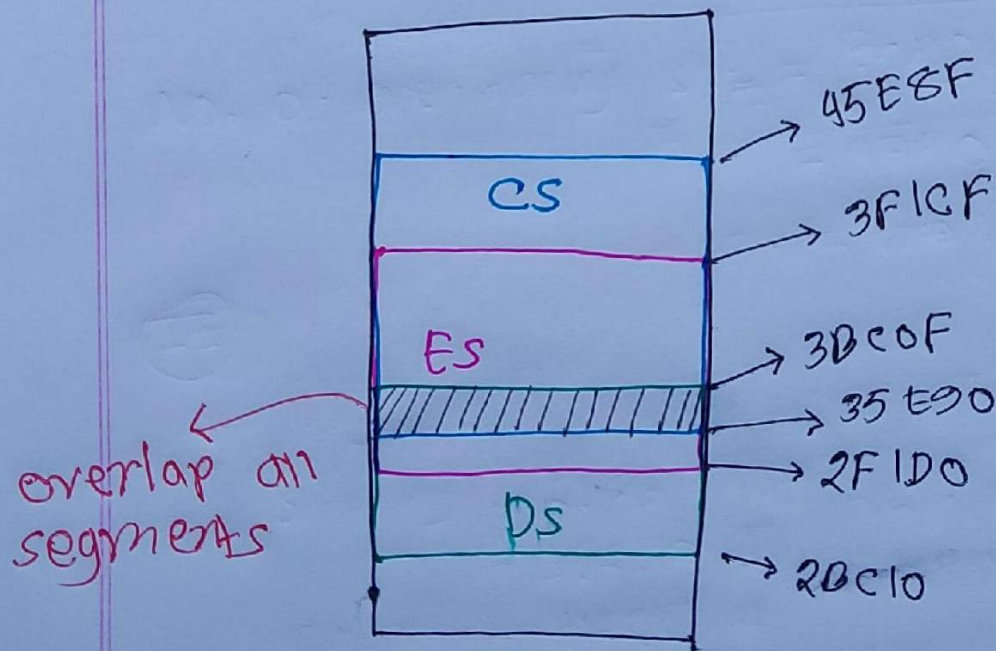
$$\begin{aligned}\text{start} &: 2BC1 \times 10H \\ &= 2BC10H\end{aligned}$$

$$\begin{aligned}\text{end} &= 2BC10 + FFFF \\ &= 3BC0F\end{aligned}$$

Es

$$\begin{aligned}\text{start} &: 2F1D \times 10H \\ &= 2F1D0\end{aligned}$$

$$\begin{aligned}\text{end} &: 2F1D0 + FFFF \\ &= 3F1CFH\end{aligned}$$



$$\begin{aligned}\text{So, No. of overlapping slots} &= 3BC0F - 35E90 \\ &= 5D7FH \\ &= 23935 \text{ (Decimal)}\end{aligned}$$